

Caleb Li

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EDUCATION

Purdue University

Bachelor of Science, Double Major: Computer Science and Math

Relevant Coursework:

Data Structures and Algorithms, C Programming, Object-Oriented Programming, Linear Algebra, Competitive Programming, Computer Architecture, Statistical Machine Learning, Artificial Intelligence, Analysis of Algorithms, Machine Learning Theory

West Lafayette, IN

May 2027 | **GPA: 3.92**

EXPERIENCE

Incoming Software Engineer Intern

Visa

May 2026 – August 2026

Highlands Ranch, CO

Undergraduate Teaching Assistant

Purdue University - Department of Computer Science

January 2026 – Present

West Lafayette, IN

- Led weekly labs and office hours for a ~**600**-student C programming course, providing hands-on debugging support and exam preparation guidance
- Mentored students on core computer science concepts, breaking down topics like data structures and memory management
- Managed online course forum, resolving **20+** technical inquiries weekly

Software Developer

Volunteer Partnerships for West Africa

October 2025 – Present

West Lafayette, IN

- Built and developed a full-stack learning management system for **200+** students in Ghana
- Architected a **RESTful API** using **Node.js** and **Express** to handle high-volume requests for course content, user data, and assignment submissions
- Built a dynamic **React** frontend with custom admin dashboards, student and teacher views, and submission windows
- Implemented **JWT Authentication** to manage role-based access control for admins, instructors, and students

AI Research Engineer (<https://deepquery.org>)

Captain (YC'W26)

September 2025 – Present

West Lafayette, IN

- Building a full-stack RAG evaluation app with **React** and **Next.js** frontend and **FastAPI** backend, deployed via **Railway**
- Implemented and benchmarked RAG pipelines/techniques including **VanillaRAG**, **RerankerRAG**, **Fusion/Iterative Retrieval**, and **Elasticsearch**
- Created synthetic datasets and leveraged **LLM-as-a-Judge** systems to evaluate RAG performance on multiple hosted datasets
- Architected a scalable data pipeline processing **1000+** Arxiv papers and **23M**-word Wikipedia dataset

Software Engineer Intern

Tencent Americas

June 2025 – August 2025

Palo Alto, CA

- Developed a full-stack web application using **Flask**, **React**, **MongoDB** and **Docker Compose** to process internal Knowledge Base and generate FAQs, improving chatbot efficiency by **50%**
- Implemented **JWT Authentication**, asynchronous processing with **Celery** and **Redis**, and concurrent user support with **Gunicorn**
- Interfaced with in-house **LLM** and **Vector Database** to build an end-to-end FAQ regression testing pipeline in Python
- Wrote Python scripts for **15** SaaS applications to scrape organization user data, clean redundant entries, and generate weekly reports; handled **100k+** lines of data; automated with **n8n**

PROJECTS

PyroTech | *Agent Orchestration, Google Gemini, Next.js, React* | https://github.com/alexyan06/Pyro_Tech

- Developed an interactive, real-time wildfire incident command simulation platform with **JavaScript/TypeScript**
- Implemented a real-time simulation engine in **Node.js** coordinating a **500ms** physics tick with **30**-minute agent cycles
- Built a **multi-agent** orchestration layer of specialized **Google Gemini 2.5 Flash** agents (Fire Behavior, Evacuation, Resources, etc.)
- Designed a physics-informed wildfire spread engine computing 72-sector elliptical perimeters per tick with head/flank/back-fire ratios modulated by wind, humidity, and temperature

re-dub | *Modal, Cloudflare (R2/D1), FastAPI, React* | <https://github.com/cli05/re-dub>

- Developed a **React** and **FastAPI** web app for automated video translation and lip-syncing, achieving a **33%** reduction in compute cost (\$/min) compared to industry benchmarks
- Architected an end-to-end video processing pipeline with **Modal**, hosting **whisper large-v3** for transcription, **llama-3.3-70b** for translation, **Coqui-XTTS** for text-to-speech, and **MuseTalk** for lip synchronization
- Engineered a speaker-adaptive pipeline in **PyTorch**, improving voice cloning similarity by **15-20%** over zero-shot methods
- Integrated **Cloudflare D1 and R2** for edge-native database and object storage functionality

TECHNICAL SKILLS

Languages: Java, Python, C++, C, TypeScript, LaTeX, HTML/CSS, CUDA, Triton

Libraries: pandas, NumPy, Matplotlib, scikit-learn, PyTorch, TensorFlow, Keras

AI/ML: Deep Learning, Diffusion/Human Motion Generation, Agents, MCP, Prompt Engineering, LangChain/Graph, GenAI